

المملكة العربية السعودية

وزارة التعليم

جامعة الإمام عبد الرحمن بن فيصل

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Designing and Simulating a Driving School

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1.Introduction

Recently, the Internet has become an important thing and a large part of our lives, so it has several types, and among these types is the creation of a network for government and private centers and facilities, and (IOT) sensors have become an important and essential thing in facilities as it connects directly to the network via smartphone, computer, or control devices connected to the World

Wide Web (WWW)[1]. In this paper, a driving school network model is simulated by Cisco Packet Tracer. The network in this model enables data storage and control of sensor devices via a smartphone. The network model contains a LAN for several departments: administration, students, guests, security, services. It also addresses security concerns by detecting smoke in case of fire and opening windows automatically.

2.Methods

2.1 Subnet: We have 5 sections and according to the law we are studying the closest number is 8, applying the law: $\text{Log}(8)=3$, $2^5 - 2=30$ Host for each subnet So, the IP used by class C starts at 192.168.1.1.

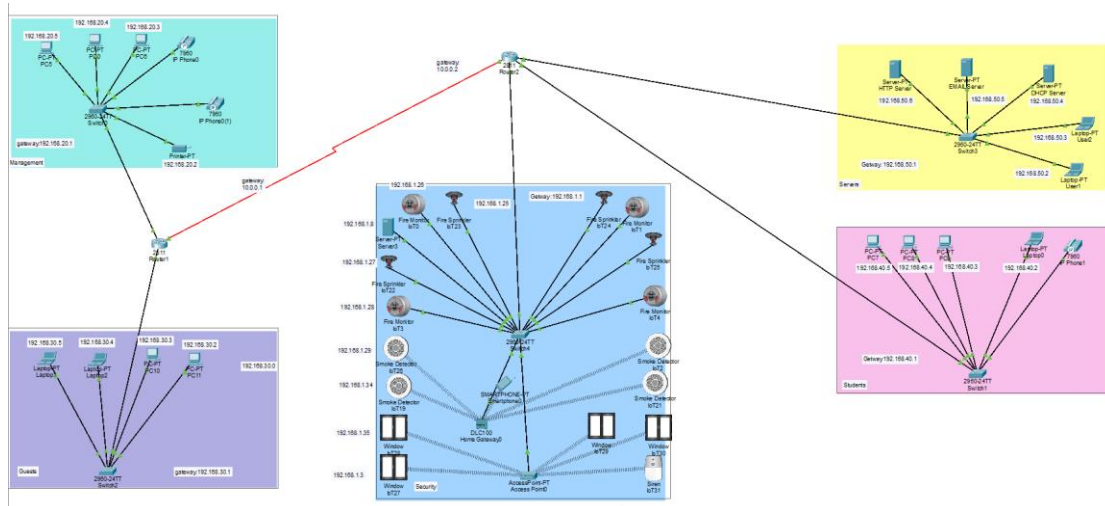


Figure (1): Sections with devices

2.2 Management:

Devices Connected to the Network[2]:

Computers (PCs): There are four computers labeled PC0[3], PC5[4], PC6[5], all connected to the router through a switch, Each computer is assigned a unique IP address within the range 192.168.1.x.

Printer: A printer is connected to the network with the IP address 192.168.1.41 , IP

Phones: Two IP phones (7960 and 7960 (1)) are included for communication purposes within the network.

Router: Gateway: The management network uses the gateway address 192.168.20.1, which indicates that it is assigned a separate network segment.

Switch: A 2960-24TT Switch acts as a central hub for connecting all devices in the management section.

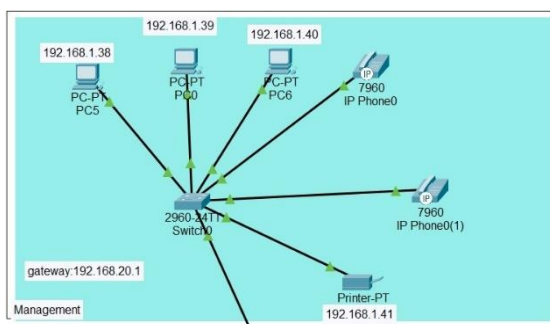


Figure (2): Management Section

```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.39

Pinging 192.168.1.39 with 32 bytes of data:

Reply from 192.168.1.39: bytes=32 time=5ms TTL=128
Reply from 192.168.1.39: bytes=32 time=10ms TTL=128
Reply from 192.168.1.39: bytes=32 time<1ms TTL=128
Reply from 192.168.1.39: bytes=32 time=7ms TTL=128

Ping statistics for 192.168.1.39:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 5ms
C:\>
```

Figure (3): PC0

```
PC5
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.38

Pinging 192.168.1.38 with 32 bytes of data:

Reply from 192.168.1.38: bytes=32 time=12ms TTL=128
Reply from 192.168.1.38: bytes=32 time<1ms TTL=128
Reply from 192.168.1.38: bytes=32 time=11ms TTL=128
Reply from 192.168.1.38: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.38:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 5ms
C:\>
```

Figure (4): PC5

```
PC6
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.40

Pinging 192.168.1.40 with 32 bytes of data:

Reply from 192.168.1.40: bytes=32 time<1ms TTL=128
Reply from 192.168.1.40: bytes=32 time<1ms TTL=128
Reply from 192.168.1.40: bytes=32 time=9ms TTL=128
Reply from 192.168.1.40: bytes=32 time=8ms TTL=128

Ping statistics for 192.168.1.40:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 9ms, Average = 4ms
C:\>
```

Figure (5): PC6

2.3 Servers:

Devices Connected to the Network:

The network consists of several devices interconnected through a 2960-24TT Switch. Below is a detailed description of the devices and their roles:

- **Servers:** There are three servers connected to the switch, each assigned a unique IP address. The HTTP Server is assigned the IP address 192.168.50.6 and is responsible for providing web services to the network. The EMAIL Server has the IP address 192.168.50.5 and manages email communications. The DHCP Server, with the IP address 192.168.50.4, dynamically assigns IP addresses to devices within the network.
- **Laptops:** Two laptops, User1 Laptop and User2 Laptop, are part of the network. User1 Laptop has the IP address 192.168.50.2, while User2 Laptop uses 192.168.50.3. Both devices are utilized for accessing network resources and services.
- **Router (Gateway):** The router, serving as the gateway for the network, is assigned the IP address 192.168.50.1. It enables communication between the internal network and external networks.
- **Switch:** A 2960-24TT Switch serves as the central hub, connecting all devices within the network and ensuring smooth data transmission between them.

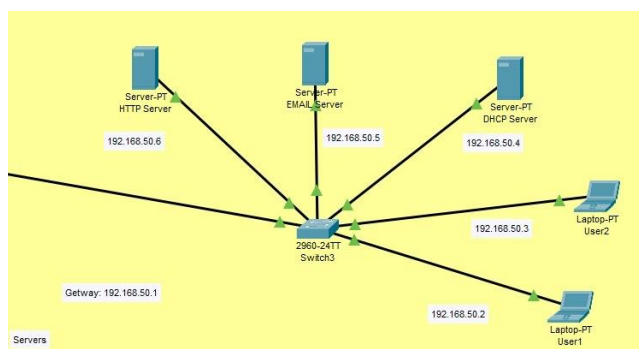


Figure (6): Servers Section

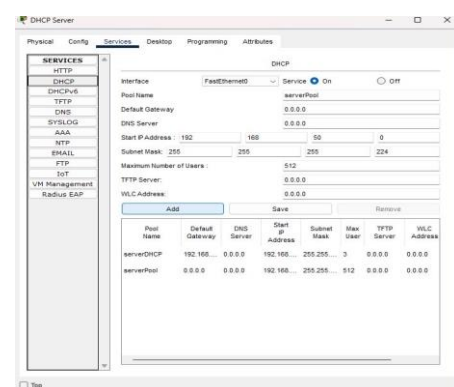


Figure (7): DHCP

2.4 Guests:

In the guest section, there are two PCs and two laptops connected to a 2960 switch. This switch is linked to router 1. The gateway for the guest network is 192.168.30.1. All devices, including the PCs and laptops, are connected through this switch, ensuring proper communication within the network as shown in Figure(7).

A ping test was conducted from PC10 (192.168.30.3) in the guest section to PC6 (192.168.20.3) in the management section, resulting in 4 replies and 0 packet loss as shown in Figure(8). Another test was performed from Laptop 1 (192.168.30.5) in the guest section to PC5 (192.168.20.5) in the management section, also showing 4 replies and 0 packet loss as shown in Figure(9).

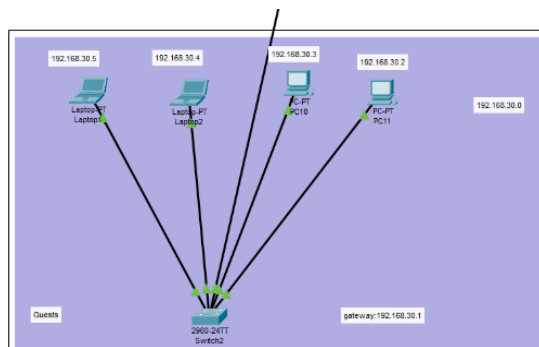


Figure (7): Guests Section

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.3

Pinging 192.168.20.3 with 32 bytes of data:

Reply from 192.168.20.3: bytes=32 time<1ms TTL=127
Reply from 192.168.20.3: bytes=32 time<1ms TTL=127
Reply from 192.168.20.3: bytes=32 time<1ms TTL=127
Reply from 192.168.20.3: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>
```

Figure (8)

```
C:\>ping 192.168.20.5

Pinging 192.168.20.5 with 32 bytes of data:

Reply from 192.168.20.5: bytes=32 time<1ms TTL=127
Reply from 192.168.20.5: bytes=32 time<1ms TTL=127
Reply from 192.168.20.5: bytes=32 time<1ms TTL=127
Reply from 192.168.20.5: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Figure (9)

2.5 Students:

Here in the student section, we have a type of switch 2960 connected to a router 2811 as shown in Figure (1) and the switch connects 5 devices:

- PCs (PC7, PC8 and PC9): Computers for students.
- Laptops (Laptop0): Portable computers for teaching or admin tasks.
- Phone (7960): A phone for communication with the management section.

All these devices are connected by using Copper Straight-Through

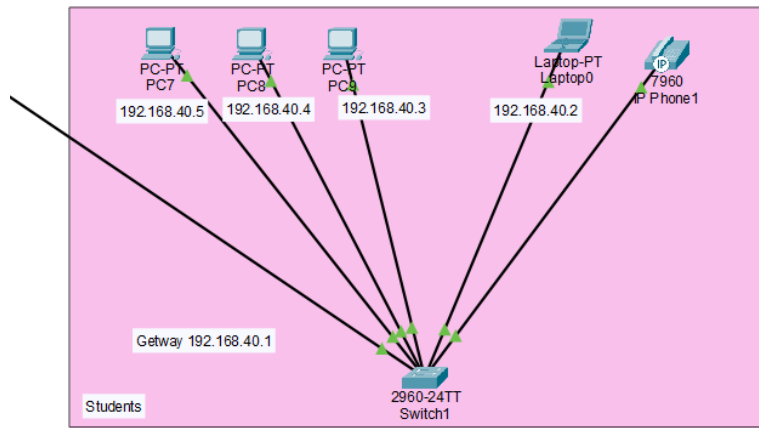


Figure (10): Students Section

2.6 Security:

In the security department, this department usually contains the following devices distributed in each department in the building as shown in the picture [11], smoke detectors monitor the atmosphere and give an alarm to the smartphone [12] so that the windows are opened to ventilate the place and operate the sprinklers, in the window sensors work automatically when smoke is detected, the importance of this department aims to protect people.

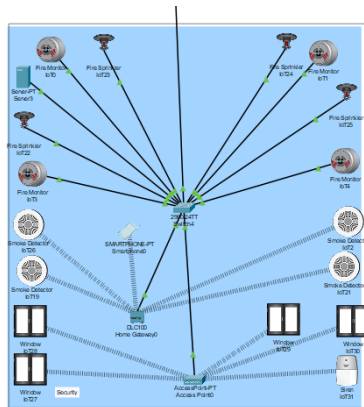


Figure (11): Security Section

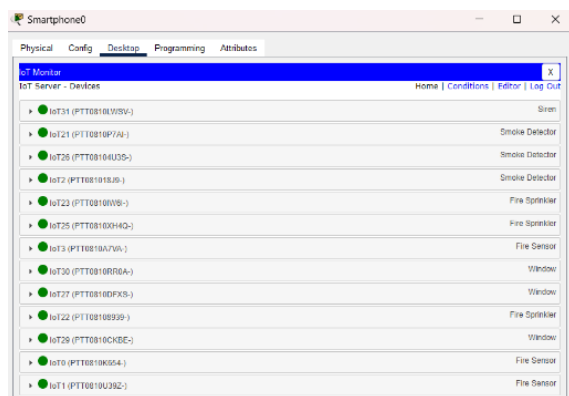


Figure (12): Security settings

When requesting access to the smartphone, only authorized persons will be allowed to enter by requesting a password and username[[]].

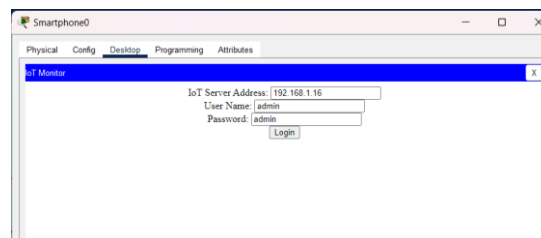


Figure (13): Login Request

3.Conclusion

In conclusion of this report, an integrated network of sections was implemented for the driving school with the aim of improving the technical structure and ensuring the provision of strong and effective communication without interruption or penetration between all sections of the school. The network included five different sections including administration, students, guests, servers, and the security system.

Modern network equipment such as switches, routers, and security sensors were used to ensure the stability and security of the network.

A picture was designed by artificial intelligence[14] of the network if it were applied on the ground.



Figure (14): Image design